MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 1: Complex numbers

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Lecture outline

1. Basic notation: logic
2. Basic notation: sets
3. Definition of the field of numbers
4. The field of complex numbers
5. Addition and subtraction of complex numbers
6. Multiplication and division of complex numbers
7. Real and imaginary part of complex number
8. Complex conjugate numbers
9. Geometry of complex plane

Basic notation: Logic. 1

Definition

If p and q are propositions, then:

- $p \vee q$ (read: p or q) denotes the disjunction of p and q. The assertion "$p \vee q$" is true whenever at least one of p,q is true.
- $p \wedge q$ (read: p and q) denotes the conjunction of p and q. The assertion "$p \wedge q$" is true only in case both p and q are true.
- $\neg\, q$ (read: not q) denotes the negation of p. The assertion "$\neg\, q$" is true only if q is false.
- $p \Rightarrow q$ is read: p implies q. By definition, "$p \Rightarrow q$" denotes "$(\neg p) \vee q$". In particular, the statement "$p \Rightarrow q$" is true if and only if the statement "$(\neg\, q) \Rightarrow (\neg\, p)$" is true.
- $p \Leftrightarrow q$ is read: p is logically equivalent to q. By definition, "$p \Leftrightarrow q$" denotes "$(p \Rightarrow q) \wedge (q \Rightarrow p)$".

Basic notation: Logic. 2

An expression $p(x)$ that becomes a proposition whenever values from a specified domain of discourse are substituted for x is called a propositional function or, equivalently, a condition on x; and p is called a property, or predicate. The assertion "y has property p" means that "$p(y)$" is true. Thus, if $p(x)$ is the proportional function "x is an integer", then p denotes the property "is an integer", and "$p(2)$" is true whereas "$p(1/2)$" is false.

The quantifier "there exists" is denoted by $\exists$, an the quantifier "for each" is denoted by $\forall$. The assertion "$\forall x\ \exists y\ \forall z : p(x, y, z)$" reads "For each x there exists a y such that for each z, $p(x, y, z)$ is true".

Basic notation: Sets. 1

Intuitively, we think of a set as something made up by all the objects that satisfy some given conditions, such as the set of prime numbers, the set of points on a line, or the set of objects named in a given list. The objects making up the set are called the elements, or members, of the set and may themselves be sets, as in the set of all lines in the plane.

Rigorously, the word *set* is an undefined term in mathematics, so that definite axioms are required to govern the use of this term. Although we shall deal with sets on an intuitive basis until we discuss an axiom systems, whenever we apply the label *set* to something, we shall later provide this usage to have been formally justified.

The membership relation is denoted by $\in$ and sets are generally indicated by capital letters: "$a \in A$" is read "a belongs to (is a member, element, point of) the set A"; $\neg(a \in A)$ is written $a \notin A$. The notation $a = b$ will mean that the objects a and b are the same, and $a \neq b$ denotes $\neg(a = b)$. If A, B sets, then $A = B$ will indicate that A and B have the same elements; that is, $\forall x : (x \in A) \Leftrightarrow (x \in B)$; $\neg(A = B)$ is written $A \neq B$.

Basic notation: Sets. 2

$A \subset B$ (or $B \supset A$), read "A is a subset of (is contained in) B", signifies that each element of A is an element of B, that is $\forall x : (x \in A) \Rightarrow (x \in B)$; equality is *not* excluded - we call A a *proper* subset of B ($A \subsetneq B$ whenever $(A \subset B) \wedge (A \neq B)$. The relations $\subset$ and $\subsetneq$ are called *inclusion* and *proper inclusion*, respectively. the following statements are evident:

1. $A \subset A$ for each set A.
2. If $A \subset B$ and $B \subset C$, then $A \subset C$ (that is, C is transitive).
3. $A = B$ if and only if both $A \subset B$ and $B \subset A$.

Of these, the last statement is very important: the equality of two sets is usually proved by showing each of the two inclusions valid.

Basic notation: Sets. 3

The axioms of set theory allow only two methods for forming subsets of a given set. One of these is by appeal to the axiom of choice. The other is by use of the following principle: If A is a set and p is a property that each element of A either has or does not have, then all the $x \in A$ having the property p from a set. This subset of A is denoted by $\{x \in A \mid p(x)\}$; it is uniquely determined by the property p. Clearly, $\{x \in A \mid p(x)\} \subset \{x \in A \mid q(x)\}$ if and only if $\forall x \in A : p(x) \Rightarrow q(x)$; thus two properties determine the same subset of A whenever each object in A having one of them also has the other.

Example

If A is the real line, the closed unit interval is $\{x \in A \mid 0 \leqslant x \leqslant 1\}$

Example

If A is the real line, $\{x \in A \mid x^2 = 1\} = \{x \in A \mid x^4 = 1\}$ even thought the defining properties are different. Note that if A was the set of complex numbers, these two properties would not determine the same subset.

Example

For each set A, $\{x \in A \mid x = x\} = \{x \in A \mid x \in A\} = A$

Elementary set theory : Field of numbers 1.

Definition

A *field* $\mathbb{F}$ is a set of elements, at least two in number, together with two binary operations known as addition $(+)$ and multiplication $(\cdot)$ subject to the following postulates:

A_1. To every ordered pair (a, b) of elements of $\mathbb{F}$ there corresponds a uniquely defined element $a + b$ in $\mathbb{F}$, called the *sum* of a and b. (closure under $+$)

A_2. $a + b = b + a$ (Commutative Law).

A_3. $(a + b) + c = a + (b + c)$ (Associative Law).

A_4. $a + c = b + c \implies a = b$ (Cancellation Law).

A_5. For every pair a, b of elements in $\mathbb{F}$ exists an element x in $\mathbb{F} \ni a + x = b$.

M_1. To every ordered pair (a, b) of elements of $\mathbb{F}$ there corresponds a uniquely defined element ab (or $a \cdot b$) of $\mathbb{F}$, called the *product* of a and b. (closure)

M_2. $\forall a, b \in \mathbb{F}$ $ab = ba$ (Commutative Law)

M_3. $\forall a, b, c \in \mathbb{F}$ $(ab)c = a(bc)$ (Associative Law)

M_4. $ac = bc$, with $c \neq 0 \implies a = b$

M_5. For every pair (a, b) of elements of $\mathbb{F}$ with $a \neq 0$, $b \neq 0$ $\exists$ and element $x \in \mathbb{F}$, $x \neq 0$, $\ni ax = b$.

D. $\forall$ $a, b, c, \in \mathbb{F}$, $a(b + c) = ab + ac$ (Distributive Law)

Elementary set theory : Field of numbers 2.

Some consequences of A_1 to A_5 are:

1. $\exists$ a unique zero element, $0 \in \mathbb{F}$ such that $a + 0 = a$, $\forall a \in \mathbb{F}$. (Identity with respect to addition)
2. For every $a \in \mathbb{F}$ $\exists$ a unique element $-a \in \mathbb{F}$, called the negative of a, such that $a + (-a) = 0$. (Additive inverse)

Some consequences of M_1 to M_5 are:

1. $\exists$ a unique unity element with respect to multiplication, denoted by 1, such that $a \cdot 1 = a \ \forall \ a \in \mathbb{F}$. (Identity with respect to multiplication)
2. For every $a \in \mathbb{F}$ such that $a \neq 0$ $\exists$ a unique inverse element with respect to multiplication, denoted by a^{-1} (called the reciprocal), and $a \cdot a^{-1} = 1$. (Multiplicative inverse)

Remark: The set of real numbers, $\mathbb{R}$, constitutes a *field* under the usual rules of addition and multiplication.

The field of complex numbers

We define $\mathbb{C}$, the complex numbers, to be the set of all ordered pairs (a, b) where a and b are real numbers and where addition and multiplication are defined by:

$$\begin{aligned}(a,b)+(c,d) &= (a+c, b+d)\\ (a,b)(c,d) &= (ac-bd, bc+ad)\end{aligned}$$

With these definitions $\mathbb{C}$ satisfies all the axioms for a field. That is $\mathbb{C}$ satisfies the associative, commutative and distributive laws for addition and multiplication: $(0,0)$ and $(1,0)$ are identities for addition and multiplication respectively, and there are additive and multiplicative inverses for each nonzero element in $\mathbb{C}$.

The field of complex numbers

We will write a for the complex number $(a, 0)$. This mapping $a \to (a, 0)$ defines a field isomorphism of $\mathbb{R}$ into $\mathbb{C}$ so we may consider $\mathbb{R}$ as a subset of $\mathbb{C}$. If we put $i = (0, 1)$ then $(a, b) = a + ib$.
Note that $i^2 = -1$, so that the equation $z^2 + 1 = 0$ has a root in $\mathbb{C}$,

$$z^2 + 1 = (z + 1)(z - 1)$$

More generally, if z and w are complex numbers we obtain

$$z^2 + w^2 = (z + iw)(z - iw)$$

The field of complex numbers

In this notation we can rewrite the formulas for the addition and multiplication of two complex numbers $z_1 = a + ib$ and $z_2 = c + id$ as

$$\begin{aligned} z_1 + z_2 &= (a + c) + i(b + d) \\ z_1 z_2 &= (ac - bd) + i(bc + ad) \end{aligned}$$

By letting a and b be real numbers $a \neq 0$ and $b \neq 0$ we can obtain

$$\frac{1}{a + ib} = \frac{a - ib}{a^2 + b^2} = \frac{a}{a^2 + b^2} - i\left(\frac{b}{a^2 + b^2}\right)$$

so that we have a formula for the reciprocal of a complex number.

The field of complex numbers

When we write $z = a + ib$ $(a, b \in \mathbb{R})$ we call a and b the *real* and *imaginary parts* of z and denote this by $a = Re\, z$, $b = Im\, z$.
We introduce the two operations on $\mathbb{C}$ which are not field operations. If $z = x + iy$ $(x,\, y \in \mathbb{R})$ than we define $|z| = (x^2 + y^2)^{\frac{1}{2}}$ to be the *absolute value* of z and $\bar{z} = x - iy$ is the *conjugate* of z. Note that

$$|z|^2 = z\bar{z}$$

In particular, if $z \neq 0$ then

$$\frac{1}{z} = \frac{\bar{z}}{|z|^2}$$

Complex plane 1

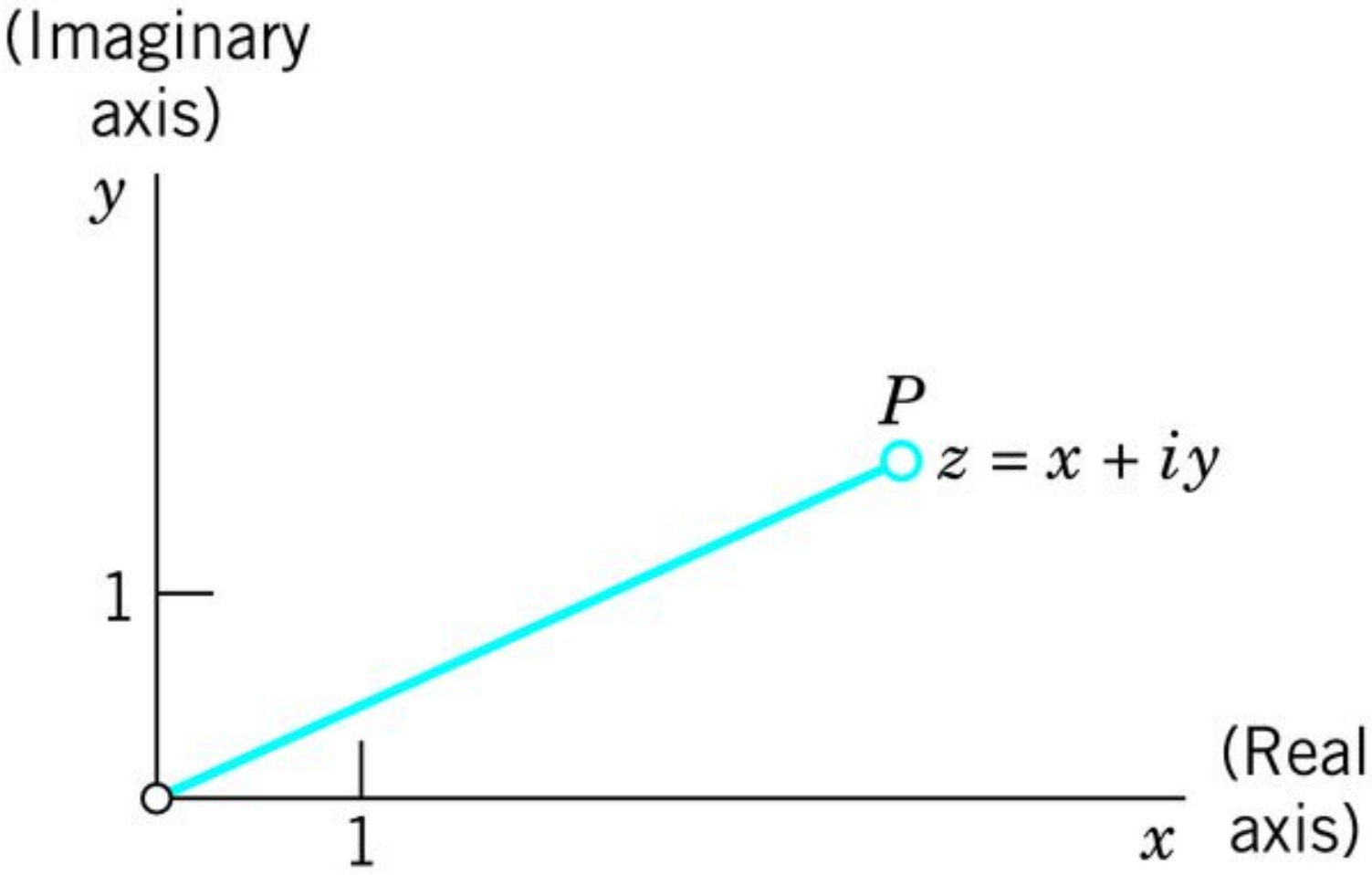

Figure: The complex plane

Complex plane 2

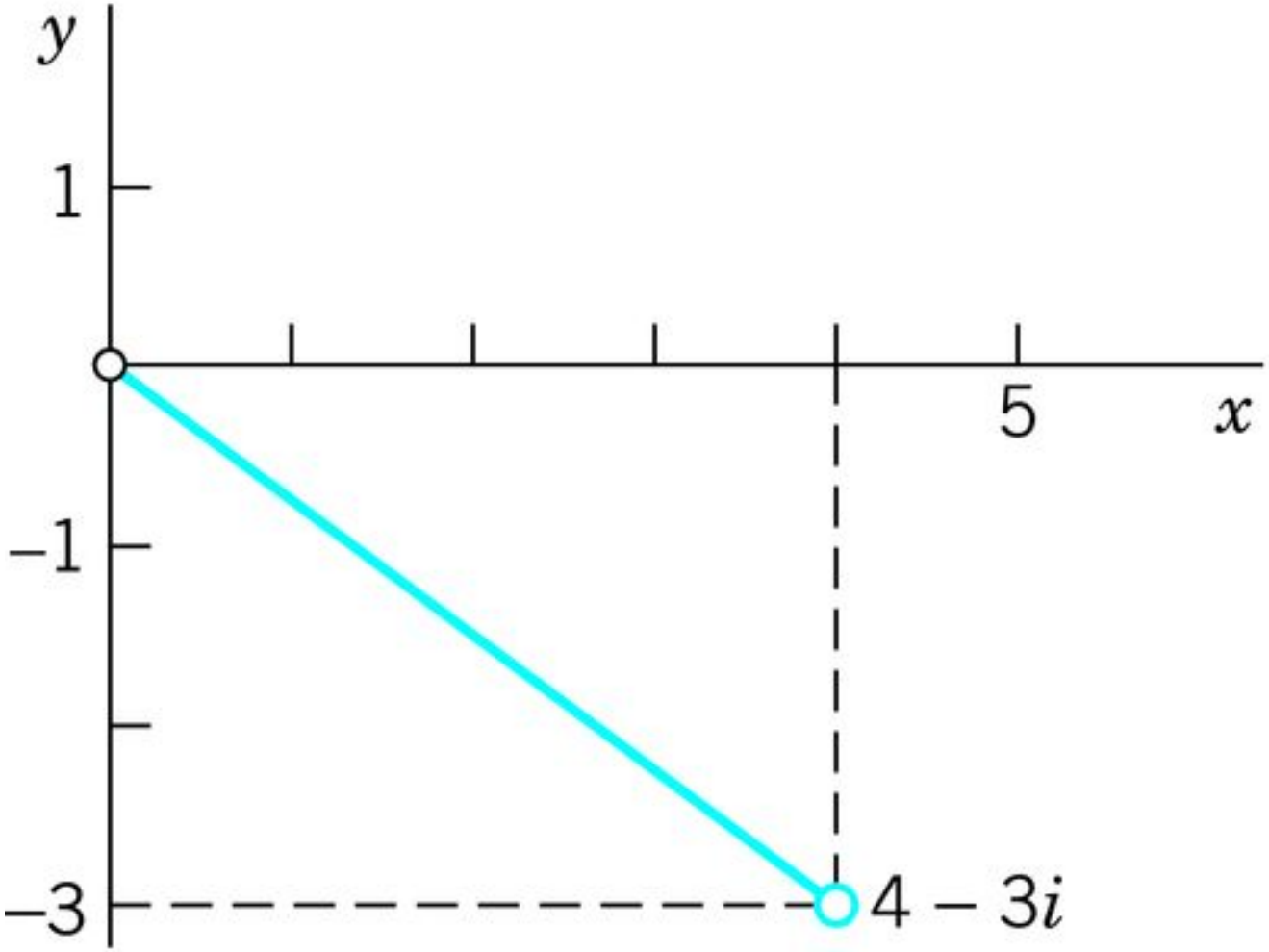

Figure: The number $4 - 3i$ in the complex plane

Complex plane 3

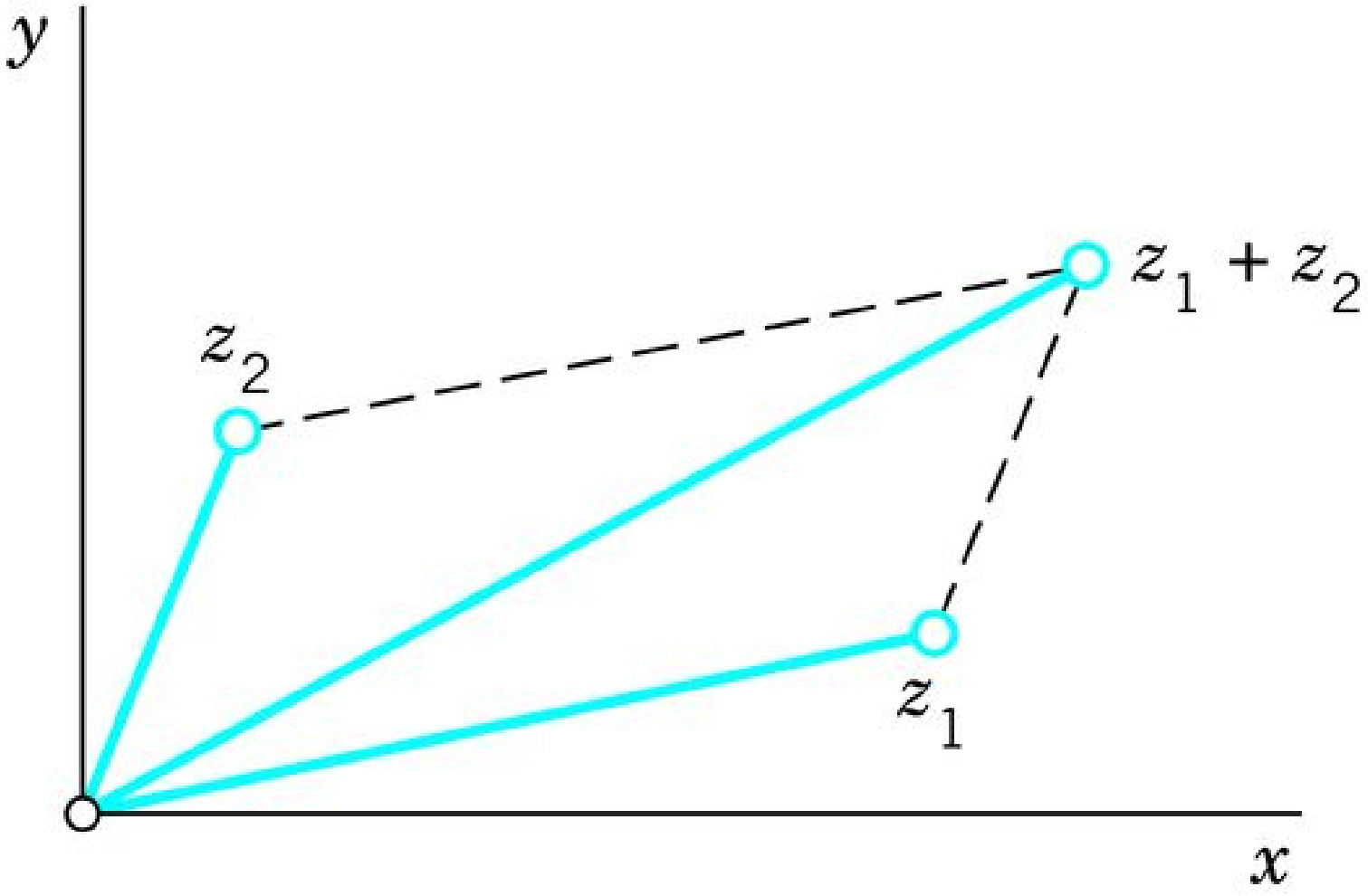

Figure: Addition of complex numbers

Complex plane 4

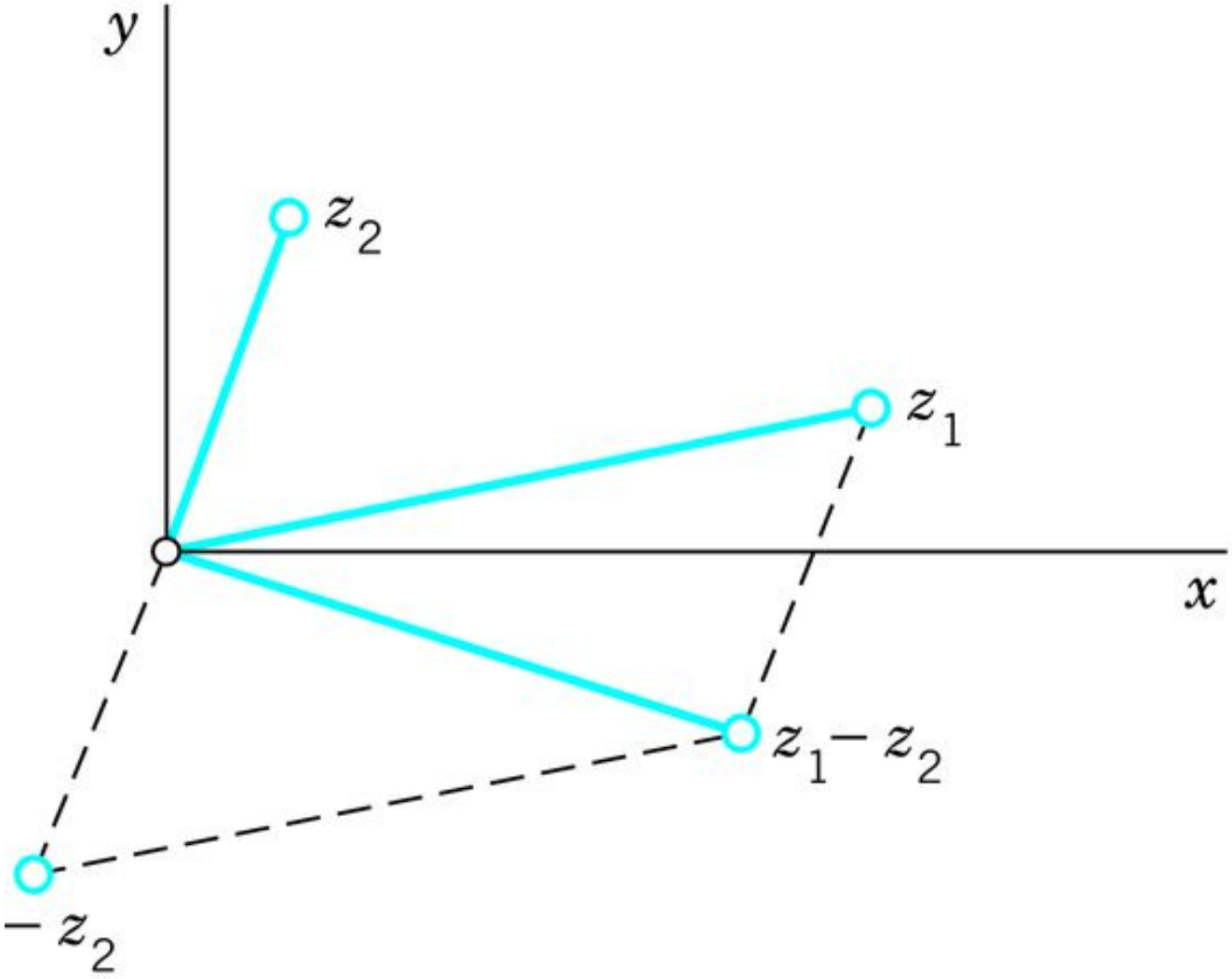

Figure: Subtraction of complex numbers

Complex plane 5

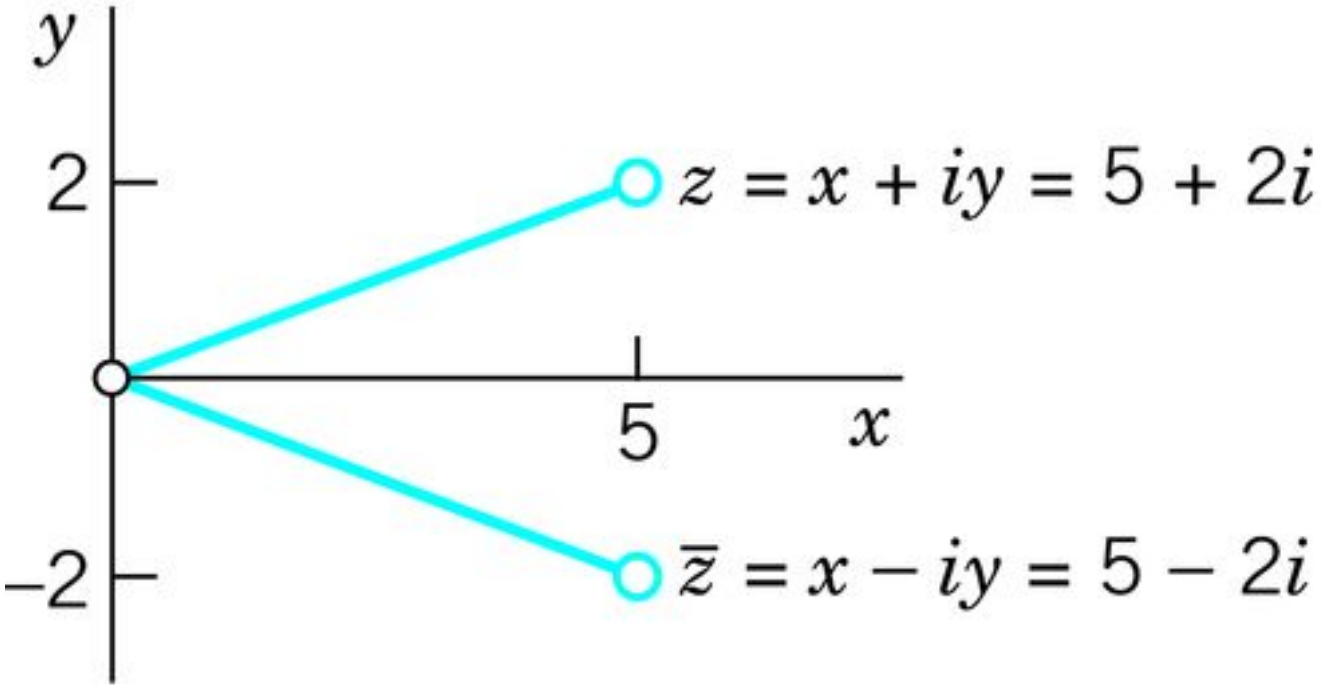

Figure: Complex conjugate numbers

Thank you